Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years of experience building **web products** across **TypeScript, Node.js, Python, React, and SQL**, shipping polished user-facing workflows, internal tooling, and AI-backed product systems with strong ownership of performance, reliability, and iteration quality.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built multi-tenant product features across **FastAPI**, **Azure SQL**, **PostgreSQL**, and cloud storage for 500+ users, reducing dashboard latency from 1.2s to 350ms through schema, indexing, and query optimization.
- Implemented background-job and integration workflows for 5K+ monthly tasks with validation, reconciliation checks, and release safeguards, improving throughput 3x while sustaining 99%+ production reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed user-facing AI grading workflows with **Flask** + **React**, secure file ingestion, and modular APIs, handling 10K+ daily requests for 150+ instructors in production.
- Built concurrent processing and rubric-generation flows on **AWS RDS** with CI/CD-backed deployments, cutting turnaround time by 60% while maintaining stable API performance under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered microservice-based APIs and data workflows using **Python**, OOP patterns, and **AWS Glue/S3**, improving response times by 30% and pipeline efficiency by 68% in a production codebase.
- Maintained 20+ unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, strengthening debugging, code review, and release quality in Agile delivery cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with **REST APIs**, WebSockets, and microservices, supporting 8+ business workflows and 2M+ synchronized records in a user-facing system.
- Implemented modular auth, billing, and notification services with **Jenkins**, **Docker**, and Azure SQL-backed workflows, reducing deployment time by 50% and sustaining 95%+ successful build validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs and real-time synchronization flows for a service-booking platform across 50K+ records, covering booking, reservation tracking, pricing, coupons, and payment status in a customer-facing product.
- Mapped 12+ integration points and implemented payment webhooks plus Firebase-backed workflows, aligning technical decisions with business requirements and production support needs.

Skills

Languages: TypeScript/JavaScript, Python, Node.js, Java, SQL, C++
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Microservices
Frontend: React.js, Next.js, TailwindCSS, HTML/CSS, WebSockets, Component Architecture, State Management
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Data Modeling, Elasticsearch Concepts
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
AI/ML: LLM Systems, AI Coding Assistants, Prompt Engineering, Codex/Claude Code Workflows, Intelligent Automation
Engineering Practices: CI/CD, Unit Testing, Integration Testing, Performance Optimization, Reliability, Internal Tooling, Product Iteration, Code Review

Projects

PiXMod | *Next.js, TailwindCSS, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS product with a responsive **Next.js** interface, image-processing workflows, and scalable storage in MongoDB, increasing premium conversion by 18% while shipping polished web experiences.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-assisted data workspace across a **Java** + **TypeScript** codebase, supporting 10+ databases and accelerating schema exploration, query authoring, and developer-facing data workflows.

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an AI document product with semantic retrieval, summarization, and real-time collaboration across 50K+ documents, improving retrieval latency by 40% through optimized indexing and data-ingestion pipelines.